Program 1

```
import java.util.Scanner;

public class Program1 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int a = sc.nextInt(), b = sc.nextInt();
    System.out.println("Addition: " + (a + b));
    System.out.println("Subtraction: " + (a - b));
    System.out.println("Multiplication: " + (a * b));
    System.out.println("Division: " + (a / b));
    System.out.println("Modulo: " + (a % b));
  }
}
```

Program 2

```
import java.util.Scanner;

public class Program2 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    double num = sc.nextDouble();
    System.out.println("Square root: " + Math.sqrt(num));
  }
}
```

Program 3

```
import java.util.Scanner;

public class Program3 {
  public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt(), b = sc.nextInt();
        int greater = (a > b) ? a : b;
        System.out.println("Greater number: " + greater);
    }
}
```

Program 4

```
import java.util.Scanner;

public class Program4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double radius = sc.nextDouble();
        double area = Math.PI * radius * radius;
        System.out.println("Area of circle: " + area);
    }
}
```

Program 5

```
import java.util.Scanner;

public class Program5 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();
        int second;
        if ((a > b && a < c) || (a > c && a < b)) second = a;
        else if ((b > a && b < c) || (b > c && b < a)) second = b;
        else second = c;
        System.out.println("Second largest: " + second);
```

```
    }
}
```

Program 6

```
import java.util.Scanner;

public class Program6 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int year = sc.nextInt();
        boolean leap = (year % 400 == 0) || (year % 4 == 0 && year % 100 != 0);
        System.out.println(leap ? "Leap Year" : "Not a Leap Year");
    }
}
```

Program 7

```
import java.util.Scanner;

public class Program7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double a = sc.nextDouble(), b = sc.nextDouble(), c = sc.nextDouble();
        double s = (a + b + c) / 2;
        double area = Math.sqrt(s * (s - a) * (s - b) * (s - c));
        System.out.println("Area of scalene triangle: " + area);
    }
}
```

Program 8

```
import java.util.Scanner;

public class Program8 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.println("Enter two numbers and operator (+ - * / %):");
    int a = sc.nextInt(), b = sc.nextInt();
    char op = sc.next().charAt(0);
    switch (op) {
      case '+': System.out.println("Result: " + (a + b)); break;
      case '-': System.out.println("Result: " + (a - b)); break;
      case '*': System.out.println("Result: " + (a * b)); break;
      case '/': System.out.println("Result: " + (a / b)); break;
      case '%': System.out.println("Result: " + (a % b)); break;
      default: System.out.println("Invalid operator");
    }
  }
}
```

Program 9

```
import java.util.Scanner;

public class Program9 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int num = sc.nextInt();
    int root = (int)Math.sqrt(num);
    System.out.println((root * root == num) ? "Perfect Square" : "Not a Perfect Square");
  }
}
```

Program 10

```
import java.util.Scanner;

public class Program10 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    double price = sc.nextDouble();
    double discount1 = price * 0.30;
    double finalPrice1 = price - discount1;

    double priceAfter20 = price * 0.80;
    double finalPrice2 = priceAfter20 * 0.90;
    double discount2 = price - finalPrice2;

    System.out.println("Discount by Shopkeeper 1: " + discount1);
    System.out.println("Discount by Shopkeeper 2: " + discount2);
  }
}
```

Program 11

```
import java.util.Scanner;

public class Program11 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    double l = sc.nextDouble(), g = sc.nextDouble();
    double T = 2 * Math.PI * Math.sqrt(l / g);
    System.out.println("Time period: " + T);
  }
}
```

Program 12

```
import java.util.Scanner;

public class Program12 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    double basic = sc.nextDouble();
    double da = 0.30 * basic;
    double hra = 0.15 * basic;
    double pf = 0.125 * basic;
    double gross = basic + da + hra;
    double net = gross - pf;
    System.out.println("Gross Pay: " + gross);
    System.out.println("Net Pay: " + net);
  }
}
```

Program 13

```
import java.util.Scanner;

public class Program13 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    double p = sc.nextDouble(), r = sc.nextDouble(), t = sc.nextDouble();
    double si = (p * r * t) / 100;
    double ci = p * Math.pow((1 + r / 100), t) - p;
    System.out.println("Difference between CI and SI: " + (ci - si));
  }
}
```

Program 14

```
import java.util.Scanner;


public class Program14 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int dividend = 2000;
    int nominal = 10, percent = 10, target = 3000;
    int currentShares = (dividend * 100) / (nominal * percent);
    int moreShares = target - currentShares;
    System.out.println("Current shares: " + currentShares);
    System.out.println("More shares to buy: " + moreShares);
  }
}
```

Program 15

```
import java.util.Scanner;
public class Program15 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int num = sc.nextInt();
    int swapped = (num % 10) * 10 + (num / 10);
    System.out.println("Swapped Number: " + swapped);
    if (num > swapped)
      System.out.println(num + " is greater");
    else if (swapped > num)
      System.out.println(swapped + " is greater");
    else
      System.out.println("Both are equal");
  }
}
```